GIT & GITHUB

Version Control with Git & GitHub

5. The Basics of Version Control

5.1 What is Version Control & SCM?

- Version Control: Track changes in code/files over time
- SCM (Source Code Management): System to manage code versions

5.2 What is Git vs GitHub?

- Git: Version control system (local & distributed)
- GitHub: Cloud platform to host Git repositories and collaborate

5.3 Git & VS Code Setup

1. Install Git: sudo apt install git
2. Configure Git:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

3. Install VS Code and Git extension for GUI workflow

5.4 Features of Git

- Commit history, branching, merging, staging, reverting

5.5 Why Git in DevOps?

- Enables CI/CD pipelines, collaborative development, rollback, traceability

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6. Let's Get Practical with Git & GitHub

6.1 GitHub Dashboard Explained

- Repositories, Issues, Pull Requests, Actions, Projects

6.2 Let's Work with Repository

```
git init            # Initialize local repo
git clone <url>     # Clone remote repo
```

6.3 Behind the Scenes: Version Control in Action on GitHub

- Local commits → push → GitHub remote repo → pull by collaborators

6.4 Git Clone vs Git Init

- git clone <url> → copy remote repo locally
- git init → create new repo locally

6.5 Ready, Stage, Commit: Your Git Workflow Essentials

```
git status              # Check changes
git add file.txt        # Stage file
git commit -m "message"   # Commit changes
```

6.6 Syncing with GitHub: Push, Pull, and Stay Updated

```
git push origin main      # Push local commits to GitHub
git pull origin main      # Pull latest changes from GitHub
```

6.7 Git in Action: Practice Time!

- Create repo, add file, stage, commit, push

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7. Branching & Merging in Git

7.1 What & Why Branch?

- Branching allows multiple development lines without affecting main code

7.2 Working with Branches Using Commands

```
git branch new-branch      # Create branch
git checkout new-branch    # Switch branch
git checkout -b new-branch # Create & switch
```

7.3 Merging Made Easy

```
git checkout main
git merge new-branch       # Merge changes into main
```

7.4 Merge Conflicts & Resolution

- Conflict occurs if same lines modified in multiple branches
- Resolve manually → git add file → git commit

7.5 Branch, Merge, Repeat: Your Git Workout

- Repeat branching, feature development, merging workflow for practice

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8. Collaborating on GitHub

8.1 What is Forking a Repository?

- Fork → copy someone else's repo to your GitHub account

8.2 How Pull Requests Work

- PR → propose changes to original repo, reviewed before merging

8.3 Hands-On: Pull Request Practice Challenges

- Fork → clone → branch → commit → push → open PR

8.4 Git - Rebase

```
git checkout feature-branch
git rebase main          # Apply feature branch commits on top of main
```

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9. Going Live: Deploying Projects Using GitHub

9.1 HTML Mini Crash Course (Bonus)

- Basic structure: <html>, <head>, <body>

9.2 CSS Mini Crash Course (Bonus)

- Inline, internal, external styles for design

9.3 Keep It Clean: Using .gitignore

- .gitignore → files/folders Git should ignore

```
node_modules/
*.log
.env
```

9.4 Git Best Practices

- Frequent commits, meaningful messages, pull before push, code review

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Git Commands

Command	Meaning
git init	Initialize local repository
git clone	Clone remote repository
git status	Show changed files
git add file	Stage file for commit
git commit -m "msg"	Commit staged changes
git push origin branch	Push local branch to remote
git pull origin branch	Pull remote changes
git branch	List branches
git branch name	Create new branch
git checkout branch	Switch branch
git checkout -b branch	Create & switch branch
git merge branch	Merge branch into current
git rebase branch	Reapply commits on top of branch
git log	View commit history
git remote -v	Show remote URLs
git rm file	Remove file from repo
git mv old new	Rename/move file
.gitignore	File to exclude files/folders from Git
fork	Copy repo to own GitHub account
pull request	Propose changes to remote repository